The one-dimensional Taylor tree as an asymptotic expansion to all orders

Claude, with Daniel Murfet

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Abstract

The O-form of unit 30, closing the one-dimensional monomial Taylor-tree programme: for $\xi(u) = \xi_0 + cu^m$ and every N,

$$Z_{\text{chart}}(n) = \sum_{p<N} \frac{(\beta c)^p}{p!}\,\frac{1}{2k}\,S_{\mu_p+p/2}(\xi_0)\,n^{-\mu_p} + O\big(n^{-\mu_N}\big), \qquad \mu_p = \frac{h+mp+1}{2k},$$

as $n \to \infty$. This is exactly `cor:standardintegralexp` in one dimension: the exponent set is $\Lambda^* = (h{+}1)/(2k) + (m/2k)\mathbb{N} \subseteq \Lambda(h,k)$, and each P_μ is a constant (no $\log n$, since $d = 1$) given by a fluctuation function. Zero `sorry`/`axiom`.

1 Setting

Unit 30 gives the N-term inequality with remainder $C_N n^{-\mu_N} + \sum_{p<N} \frac{(\beta|c|)^p}{p!}\mathrm{tail}_p$. Each tail is $(\sqrt{n})^p$ times a plain boundary tail, hence at most $K_p\, n^{p/2}e^{-\varepsilon n}$ by unit 25, and $n^{p/2}e^{-\varepsilon n} = o(n^{-\mu_N})$. Collecting the finitely many eventual bounds gives the asymptotic statement.

2 The thing formalised

```
theorem standardIntegral1D_taylor_tree_isBigO (    c b :  ) (h k m N :  )
    (h  : 0 <  ) (hb : 0 < b) (hk : 0 < k) :
    (fun n :   => Z_chart n
        -   p   range N, (  c)^p/p! * ((1/(2k)) n^(- _p) S_{ _p+p/2}( )))
      =O[atTop] fun n :   => n ^ (- _N)
```

3 Proof strategy

As in unit 29 but uniformly in p: the little-o scale $n^{p/2}e^{-\varepsilon n} = o(n^{-\mu_N})$ is `IsBigO.mul_isLittleO` of $n^{p/2} = O(n^{p/2})$ with `isLittleO_exp_neg_mul_rpow_atTop` at $-\mu_N - p/2$, then `IsLittleO.congr'` via `rpow_add`. The N eventual bounds are gathered by `Filter.eventually_all_finset` into one `filter_upwards` together with $n \geq 1$. Each tail is rewritten as $(\sqrt{n})^p \int_b^\infty u^{h+mp+kp} e^{\mathrm{pop}}$ (`mul_pow`, `pow_mul`, `pow_add`, `integral_const_mul`), bounded by `standardIntegral1D_tail_le`, and $(\sqrt{n})^p = n^{p/2}$ by `sqrt_eq_rpow`/`rpow_natCast`/`rpow_mul`. The final sum is handled by `gcongr with p hp` termwise and `Finset.sum_mul`, producing the constant $C_N + \sum_{p<N} \frac{(\beta|c|)^p}{p!} K_p$ for `IsBigO.of_bound`.

4 Roadblocks and resolutions

None: first-try compile. The one idiom new relative to unit 29 is `Filter.eventually_all_finset`, which turns "for each $p < N$, eventually …" into a single eventual statement usable inside `filter_upwards`.

5 What was Mathlib, what was new

Mathlib: `Filter.eventually_all_finset`, `IsBigO.mul_isLittleO`, `IsLittleO.congr'`, `IsBigO.of_bound`, `Finset.sum_mul`, `Real.Gamma_pos_of_pos`. New: the complete one-dimensional Taylor-tree asymptotic expansion for a monomial perturbation, at every order, with explicit fluctuation-function coefficients.

6 Outcome

With this unit the one-dimensional, monomial-perturbation instance of grammar §4.2 is fully formalised: kernel identity (22), insertions (24), Gaussian domination (23), boundary tail (25), exact series (26), leading asymptotic (27), two-term and all-orders expansions with explicit remainders (28, 30) and their O-packagings (29, 31). What remains of §4.2 is genuinely multivariate: analytic (non-monomial) ξ via the tree combinatorics, $\log n$ powers from repeated poles when $d > 1$, and the state-density Mellin formulation.